

TNA

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```
names(data) <- gsub("^section_[a-z]-", "", names(data))
names(data) <- gsub("-", "_", names(data))
data <- as_tibble(data, .name_repair = "unique")

new_names_dots <- c(
  "SubmissionDate", "start", "end", "today", "note.0", "note.1", "date.filling",
  "directorates", "division", "gender", "age.bracket", "name", "pf.no",
  "designation", "grade", "service.length", "duty.1", "duty.2", "duty.3",
  "duty.4", "duty.5", "duty.6", "job.desc.match", "role.clear", "task.change",
  "perf.rating", "targets.met", "challenge", "other.challenge", "note.4",
  "tech.skill", "ict.skill", "comm.skill", "mgmt.skill", "policy.skill",
  "finance.skill", "cust.service", "me.skill", "improve.skill", "attended.train",
  "train.1.y", "train.2.y", "train.3.y", "train.4.y", "train.5.y", "train.6.y",
  "train.1.n", "train.2.n", "train.3.n", "train.4.n", "train.5.n", "train.6.n",
  "train.effective", "skill.acquired", "no.skill.reason", "rel.train.1",
  "rel.train.2", "rel.train.3", "rel.train.4", "rel.train.5", "rel.train.6",
  "career.train.1", "career.train.2", "career.train.3", "career.train.4",
  "career.train.5", "career.train.6", "train.need.lvl", "train.type",
  "train.mode", "duration", "train.time", "support.h", "barrier", "other.support.h",
  "trends", "other.support.i", "new.skill.1", "new.skill.2", "new.skill.3",
  "new.skill.4", "new.skill.5", "new.skill.6", "comments", "notexx", "notexxx",
  "instanceID", "KEY", "SubID", "SubName", "AttachPres", "AttachExp",
  "Status", "Review", "DeviceID", "Edits", "Version"
)

# Apply to your data
colnames(data) <- new_names_dots

# Check the result
head(names(data), 10)
```

```
[1] "SubmissionDate" "start"           "end"             "today"
```

```
[5] "note.0"          "note.1"          "date.filling"    "directorate"
[9] "division"        "gender"
```

```
library(viridis)
```

Loading required package: viridisLite

```
library(hrbrthemes) # For clean typography
```

Warning: package 'hrbrthemes' was built under R version 4.5.3

```
# 1. Prepare and Score Data using your .dot names
directorate.skills <- data %>%
  mutate(across(c(tech.skill, ict.skill, comm.skill, mgmt.skill,
                  policy.skill, finance.skill, cust.service, me.skill),
    ~case_when(
      . == "high" ~ 3,
      . == "medium" ~ 2,
      . == "low" ~ 1,
      TRUE ~ NA_real_
    ))) %>%
  select(directorate, tech.skill, ict.skill, comm.skill, mgmt.skill,
    policy.skill, finance.skill, cust.service, me.skill) %>%
  pivot_longer(cols = -directorate, names_to = "Skill.Area", values_to = "Rating") %>%
  group_by(directorate, Skill.Area) %>%
  summarise(Average.Rating = mean(Rating, na.rm = TRUE), .groups = "drop") %>%
  # Clean up labels: remove dots, capitalize
  mutate(Skill.Area = gsub("\\.", " ", Skill.Area) %>% str_to_title())

# 2. Level 8 Plot - Standardized for Windows Compatibility
ggplot(directorate.skills, aes(x = Skill.Area, y = reorder(directorate, Average.Rating), fill = Average.Rating)) +
  geom_tile(color = "white", size = 0.5) +
  # Diverging Scale: Red (1) -> Yellow (2) -> Green (3)
  scale_fill_gradient2(low = "#d73027", mid = "#fee08b", high = "#1a9850",
    midpoint = 2, limit = c(1,3),
    breaks = c(1, 2, 3), labels = c("Low", "Med", "High"),
    name = "Proficiency") +
  theme_minimal(base_family = "sans") + # Uses standard Windows 'sans' font
  theme(
    axis.text.x = element_text(angle = 35, hjust = 1, size = 10, face = "bold", color = "black"),
    axis.text.y = element_text(size = 10, face = "bold", color = "black"),
```

```

plot.title = element_text(size = 16, face = "bold"),
plot.subtitle = element_text(size = 11, color = "grey30"),
panel.grid = element_blank(),
legend.key.height = unit(1, "cm")
) +
labs(
  title = "Organizational Skill Gap Analysis 2025/26",
  subtitle = "Aggregated levels by Directorate. Red identifies training priorities.",
  x = NULL,
  y = "Directorate"
) +
# Adds numeric values inside the tiles
geom_text(aes(label = round(Average.Rating, 1)), color = "black", size = 3.5)

```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.

